

Expected multi-utility representation $\Rightarrow$ Transitivity

Result — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Premises

- **Expected multi-utility representation.** There is a set U of functions $u : X \rightarrow \mathbb{R}$ such that for all $p, q \in \Delta(X)$: $p \succcurlyeq q$ if and only if $EU_u(p) \geq EU_u(q)$ for every $u \in U$.

0.2 Conclusion

- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.

0.3 Proof

If $EU_u(p) \geq EU_u(q)$ and $EU_u(q) \geq EU_u(r)$ for every $u \in U$ then $EU_u(p) \geq EU_u(r)$ for every $u \in U$.

Record: topics/decision-theory/results/multi-utility-implies-transitivity.yaml